Lecture 8

Applications of Least Squares
- Data Fitting
- Interpolation
- Approximation
- Weighted Least Sq.

Last time we discussed least squares solutions to linear systems. Now we will see how least squares are used in applications. One of the most important applications of the least squares is data fitting.

Suppose we are studying a certain phenomenon, which can be schematically represented as follows:

$$\text{inputs } t_i \longrightarrow \boxed{\text{Nature}} \longrightarrow \text{outputs } y_i$$

t_i could be actual inputs, or simply observation times. (measurement times)

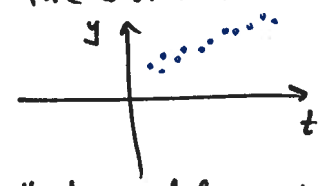
Let $(t_1, y_1), \dots, (t_m, y_m)$ be the collected data.

Suppose that the theory underlying the phenomenon indicates that $\boxed{y = \alpha + \beta t}$ theoretical model

Goal: to find α and β from the data $\{(t_i, y_i)\}$

Simple example: An object is moving with a constant speed β and at time $t=0$ its position was α.
$\Rightarrow$ at time t its position is $y = \alpha + \beta t$.

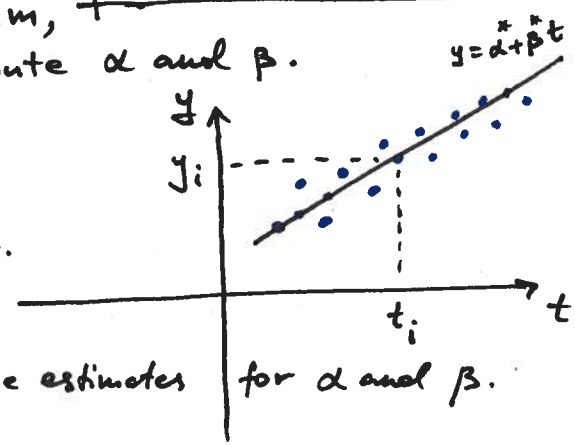
Remark: We may not have any theory for the phenomenon. We can check the scatterplot of the data and see that



So that model $y = \alpha + \beta t$ is reasonable.

In an ideal world, where the theory is absolutely correct (i.e. exactly describes the phenomenon) and there are no measurement errors, $y_i = \alpha + \beta t_i$ for all $i = 1, \dots, m$, and just two data points are enough to compute α and β.

In reality, even if the linear model is correct, the data looks like this:
(measurement errors are inevitable)
$\Rightarrow$ there is no line that goes exactly through all data point.



Goal: to find a line $y = \alpha^* + \beta^* t$ that "fits best" the measured data and use α^* and β^* as the estimates for α and β.

Question What does it mean "fits best"?

There are several ways to define the best fit, here is one:

The difference between the observed value y_i and the value predicted by the model is called the residual: $e_i = \underbrace{y_i}_{\text{measurement}} - \underbrace{(\alpha + \beta t_i)}_{\text{model prediction}}$

residual: also called prediction error in engineering fields.

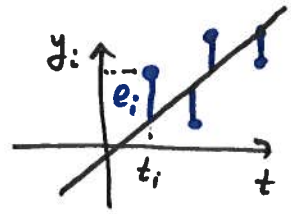
Let $e = (e_1, \dots, e_m)^T$ be the vector of residuals.

$$\begin{bmatrix} e_1 \\ \vdots \\ e_m \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$e = y - A \cdot x$$

We want all residuals to be small. The overall measure of the fit is the Euclidean norm of e.

Thus, we are looking for the coefficient vector $x^* = (\alpha^*, \beta^*)^T$ that minimizes the Euclidean norm of the residual vector:



$$(*) \qquad \boxed{x^* = \arg\min_{x \in \mathbb{R}^2} \| \overbrace{Ax - y}^{e} \|_2} = \|e\|_2$$

Geometrically:

But this is exactly the least squares solution to the system $Ax = y$.

Remark In principle, we can minimize $\|e\|_1 = |e_1| + \dots + |e_m|$ or $\|e\|_\infty = \max\{|e_1|, \dots, |e_m|\}$, but these minimization problems are much harder, nonlinear, and to solve them, we need to use tools outside of linear algebra. As a result of simplicity, least squares (*) is used in most applications.

Last time we established the following result: if $\ker A = \{0\} \Rightarrow x^* = (A^TA)^{-1}A^Ty$

In our case, $A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}$ Therefore, $\ker A = \{0\} \Leftrightarrow$ colums of A are linearly ind $\Leftrightarrow$ not all t_i are equal. (a very weak assumption)

Basically, we need to measure ys at at least two distinct times (or two distinct inputs).

Under assumptions that not all t_i's are equal:

$$A^TA = \begin{pmatrix} 1 & \dots & 1 \\ t_1 & \dots & t_m \end{pmatrix} \begin{pmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{pmatrix} = \begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} = m \begin{bmatrix} 1 & \bar{t} \\ \bar{t} & \overline{t^2} \end{bmatrix} \qquad \bar{t} = \frac{1}{m}\sum t_i, \quad \overline{t^2} = \frac{1}{m}\sum t_i^2$$

$$A^Ty = \begin{pmatrix} 1 & \dots & 1 \\ t_1 & \dots & t_m \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = \begin{bmatrix} \sum y_i \\ \sum t_i y_i \end{bmatrix} = m \begin{bmatrix} \bar{y} \\ \overline{ty} \end{bmatrix} \qquad \overline{ty} = \frac{1}{m}\sum_{i=1}^m t_i y_i$$

average (mean) values

$$\Rightarrow \quad m \begin{bmatrix} 1 & \bar{t} \\ \bar{t} & \overline{t^2} \end{bmatrix} \begin{bmatrix} \alpha^* \\ \beta^* \end{bmatrix} = m \begin{bmatrix} \bar{y} \\ \overline{ty} \end{bmatrix}$$

Test question:

1. $\overline{t^2} \neq \bar{t}^2$
2. $\overline{ty} \neq \bar{t}\,\bar{y}$

This system of two equations is easy to solve:

$$(\#) \qquad \alpha^* = \bar{y} - \bar{t}\beta^*, \quad \beta^* = \frac{\overline{ty} - \bar{t}\bar{y}}{\overline{t^2} - (\bar{t})^2} = \frac{\sum (t_i - \bar{t})(y_i - \bar{y})}{\sum (t_i - \bar{t})^2}$$

Thus, the best — in the least squares sense — straight line that fits the given data $\{(t_i, y_i)\}$, $i = 1 \dots m$ is $y = \alpha^* + \beta^* t$.

Remark Often problems that don't look like linear least squares problems can be converted to the least squares formulation by taking appropriate transformations of participating variables.

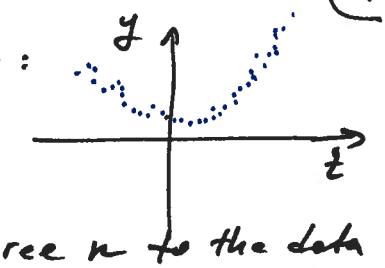
Let m_i be the (measured) amount of radioactive material in a sample of an unknown isotope at time t_i. Data: $\{(t_i, m_i)\}$.

Theory $m(t) = m_0 e^{\beta t}$, m_0 intimal mass, $\beta < 0$ decay rate. Problem: find m_0 and β

The model is not linear, but: $\log m(t) = \overbrace{\log m_0}^{\alpha} + \beta t$ we can fit this to $D = \{(t_i, \log m_i)\}$.

Polynomial Approximation and Interpolation

Suppose the scatterplot of the data $\{(t_i, y_i)\}$ looks like this:



We can fit a line to the data, but it does not really make sense.
A parabola $y = \alpha + \beta t + \gamma t^2$ seems to be a better model.
In general, suppose we want to fit a polynomial of degree n to the data

$$y(t) = \alpha_0 + \alpha_1 t + \dots + \alpha_n t^n, \quad \alpha_i \in \mathbb{R}. \qquad (\bullet)$$

The ith residual: $e_i = y_i - y(t_i) = y_i - (\alpha_0 + \alpha_1 t_i + \dots + \alpha_n t_i^n)$, $i = 1, \dots, m$

$$\underbrace{\begin{bmatrix} e_1 \\ \vdots \\ e_m \end{bmatrix}}_{e} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}}_{y} - \underbrace{\begin{bmatrix} 1 & t_1 & \dots & t_1^n \\ 1 & t_2 & \dots & t_2^n \\ \vdots & \vdots & & \\ 1 & t_m & \dots & t_m^n \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_n \end{bmatrix}}_{x}$$

A is called a Vandermonde matrix (French mathematician that did not introduce the Vandermonde matrix)

$$A \in \mathbb{M}_{m \times (n+1)}$$

<u>Consider a special case</u>: $m = n+1$ (# measurements = # coefficients).

$\Rightarrow$ A is square and, if A is nonsingular, we can find x such that $e = 0$
In other words, we can solve $Ax = y$ exactly, i.e find a polynomial $(\bullet)$ that fits the data $\{(t_i, y_i)\}$ exactly. This polynomial is called <u>interpolating polynomial</u>.

<u>Lemma</u> If $t_1 \dots t_{n+1}$ are all distinct $(t_i \neq t_j)$
$\Longleftrightarrow$ the $(n+1) \times (n+1)$ Vandermond matrix is nonsingular.

<u>Remark</u> Textbook gives a proof based on LU decomposition.
But the statement is very intuitive if you think about it geometrically

<u>Theorem 1</u> Let $t_1 \dots t_{n+1}$ be distinct sample points. Then for any $y_1 \dots y_{n+1}$
$\exists!$ interpolating polynomial of degree $\leq n$ such that $y(t_i) = y_i$ $\forall i = 1 \dots n+1$

Moreover, there is an explicit formula for the interpolating polynomial.
It is based on the <u>superposition principle</u> for inhomogenious systems (Lecture 3)

<u>Recall</u>: if $x_1^* \dots x_k^*$ are solutions of $Ax = b_1, \dots, Ax = b_k$, then $x^* = \sum_{i=1}^{k} \alpha_i x_i^*$ is a solution of $Ax = b$, where $b = \sum_{i=1}^{k} \alpha_i b_i$

We want to solve $Ax = y$ ← vandermonde (nonsing.)
Let us decompose $y = y_1 e_1 + \dots + y_{n+1} e_{n+1}$,
where $\{e_1, \dots, e_{n+1}\}$ is the standard basis in $\mathbb{R}^{n+1}$.
If we can solve

$Ax = e_k$ for all $k = 1 \dots n+1$ and x_k^* is the solution

$\Rightarrow$ our solution $x^* = y_1 x_1^* + \dots + y_{n+1} x_{n+1}^*$.

we need to find a polynomial: $= \begin{cases} 1 & \text{at } t_k \\ 0 & \text{at } t_i, i \neq k \end{cases}$ (#)

Theorem 2 Let $t_1 \dots t_{n+1}$ be distinct points. The polynomial

k^{th} Lagrange interpolating polynomial $\longrightarrow$

$$\boxed{L_k(t) = \frac{(t-t_1)\dots(t-t_{k-1})(t-t_{k+1})\dots(t-t_{n+1})}{(t_k-t_1)\dots(t_k-t_{k-1})(t_k-t_{k+1})\dots(t_k-t_{n+1})}}, \quad k=1,\dots,n+1$$

is the unique polynomial of degree n such that

x_k^* is a vector of coefficient for $L_k(t)$

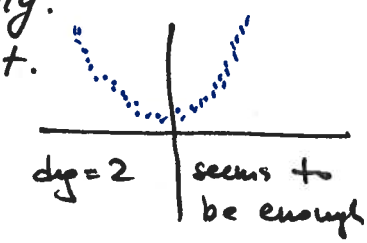
$$(\#) \quad L_k(t_i) = \begin{cases} 1 & i=k \\ 0 & i\neq k \end{cases}$$

Proof Uniquness follows from Th 1. The fact that L_k satisfies (#) is readily checkable.

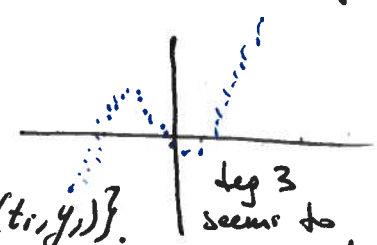
Theorem 3 Let $t_1 \dots t_{n+1}$ be distinct points. Then the polynomial of degree $\leq n$ that interpolates the data $\{(t_i, y_i)\}_{i=\overline{1,n+1}}$ is

$$\boxed{p(t) = y_1 L_1(t) + \dots + y_{n+1} L_{n+1}(t)}$$

Remark 1 If you have m data points $\{(t_i, y_i)\}$, $i=\overline{1,m}$ it is not always a good idea to fit it with the interpolating polynomial exactly. It is always usefull to examine the corresponding scatterplot. It is often the case that low degree polynomials are good theoretical models for the data.



Remark 2 Interpolating polynomials are used for approximation of complicated functions: if we want to approximate f (say for numerical integration) on $[a,b]$, we compute $y_i = f(t_i)$, $t_i \in [a,b]$, $i=\overline{1,m}$ and then approximate f with the polynomial that interpolates $\{(t_i, y_i)\}$. Intuitively, the larger m (the large the deg), the more accurate approxim. Generally, this is not true. High degree interpolating polynomials often behave badly, especially near the ends of the interval (homework). In practice, low degree polynomials are used or piecewise cubic splines.



If $m > n+1 \implies \underbrace{\begin{bmatrix} 1 & t_1 \dots & t_1^n \\ 1 & t_2 \dots & t_2^n \\ \vdots & & \\ 1 & t_m \dots & t_m^n \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_n \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}}_{y}$ does not (usually) have a solution (more equations than unknowns).

$\implies$ we must switch to a least squares approximation:

$$x^* = \arg\min \|e\|_2 = \arg\min_x \|Ax - y\|_2$$

If $\ker A = \{0\} \iff \operatorname{rank} A = n+1$ (full rank) $\iff$ at least $(n+1)$ values out of $t_1 \dots t_m$ are distinct

In this case $\implies x^* = (A^T A)^{-1} A^T y$.

Approximation & Interpolation by General Functions

There is nothing special about polynomials : we can approximate the data $\{(t_i, y_i)\}$ by a linear combination $\boxed{y(t) = \alpha_1 h_1(t) + \dots + \alpha_n h_n(t),}$ (*) where $h_1 \dots h_n$ are fixed specified functions.

Remark most general case

- Straight line case : $n=2$, $h_1(t)=1$, $h_2(t)=t$
- Polynomial case : $h_1(t)=1$, $h_2(t)=t, \dots, h_{n+1}(t)=t^n$

As long as the theoretical model (*) is linear in the parameters $\alpha_1 \dots \alpha_n$, we can always find the least squares approximation.

As before, the ith residual

$$e_i = y_i - y(t_i) = y_i - (\alpha_1 h_1(t_i) + \dots + \alpha_n h_n(t_i))$$

$$\begin{bmatrix} e_1 \\ \vdots \\ e_m \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} h_1(t_1) \dots h_n(t_1) \\ \\ h_n(t_m) \dots h_n(t_m) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

$$e = y - A \cdot x$$

- If A is square ($m=n$) and nonsingular, we can find the interpolating function of the form (*), by solving $Ax=y$ exactly.

If $m>n$ $\Rightarrow$ we can't solve $Ax=y$ exactly (in general), but we can find the least squares approx:

(#) $x^* = (A^T A)^{-1} A^T y$ (if $\ker A = \{0\}$ columns are lin. ind. rank $A = n$)

Checking $\ker\{A\} = \{0\}$ is more difficult in general case than in polynomial. (for example, lin. indep. of $h_1(t) \dots h_n(t)$ and $t_i \neq t_j$ is not enough in general), e.e. sampled vectors may still be lin. dep. (see Exercise 2.3.37)

So : if rank $A = n \Rightarrow$ (#), if rank $A < n \Rightarrow$ no closed form solution.

Remark A particularly important case :

$\{h_i\} = \{1, \cos t, \sin t, \cos 2t, \sin 2t, \dots, \cos nt, \sin nt\}$ $(2n+1)$ trigonometric functions.

$\Rightarrow$ interpolation on $(2n+1)$ equally spaced points t_i on the interval $[0, 2\pi]$ leads to the Discrete Fourier Transform, used in signal processing, data compression.

Weighted Least Squares

A useful extension to the standard least squares method is to introduce weights associated with each data point : $w_i \longleftrightarrow (t_i, y_i)$, $w_i > 0$.

The larger the weight, the more "important" the corresponding data point (for example, some of (t_i, y_i) could be known for being more accurate, reliable or significant)

Fitting the model to (t_i, y_i) with large weights is more important, the corresponding errors e_i are more vital.

So instead minimizing $\|e\|_2 = \sqrt{\sum_{i=1}^{n} e_i^2}$ (standard least squares),

we will minimize $\|e\|_w = \sqrt{\sum_{i=1}^{m} w_i e_i^2}$ (weighted least squares)

Let $W = \mathrm{diag}(w_1 \dots w_m)$. Since all $w_i > 0$, W is positive definite and $\|e\|_W = \sqrt{e^T W e}$ is the norm of the residual vector associated with the weighted inner product $\langle a, b \rangle_W = a^T W b$.

<u>Consider the case of a straight line model:</u>

$$y(t) = \alpha + \beta t.$$

(straight forward to generalize to any in param. $\rightarrow$ <u>linear</u> model)

$\Rightarrow e_i = y_i - (\alpha + \beta t_i)$ and as before $e = y - Ax$, where $y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$, $x = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

We want to minimize $\|e\|_W$, that is find $x^* = \arg\min_x \|e\|_W$

$$A = \begin{bmatrix} 1 & t_1 \\ 1 & t_m \end{bmatrix}$$

$$\|e\|_W^2 = \|y - Ax\|_W^2 = (y - Ax)^T W (y - Ax) =$$

$$= y^T W y - \underbrace{x^T A^T W y} - \underbrace{y^T W A x} + x^T A^T W A x = x^T \underbrace{A^T W A}_{K} x - 2 x^T \underbrace{A^T W y}_{f} + \underbrace{y^T W y}_{\|y\|_W^2 = c}$$

$$= x^T K x - 2 x^T f + c \longrightarrow \min$$

Everything boils down to minimization of quadratic function of x.

<u>Question</u> What is matrix K?

K is the Gram matrix associated with columns of A: $\underbrace{\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} t_1 \\ \vdots \\ t_m \end{pmatrix}}_{\text{vectors of an inner space with inner product } \langle \cdot, \cdot \rangle_W}$

We know (lecture 6) that $K > 0 \Leftrightarrow$ if the vectors K associated with are lin. independent.

$\Rightarrow$ If not all t_i are the same, then $K > 0$.

But if $K > 0$ and $K^T = K^T$, then (lecture 7) $\exists!$ minimazer which is the solution of $Kx = f$. In our case, $A^T W A x = A^T W y$

Finally: $\boxed{x^* = (A^T W A)^{-1} A^T W y}$

<u>Remark</u> If $W = I_m \Rightarrow$ standard least squares.